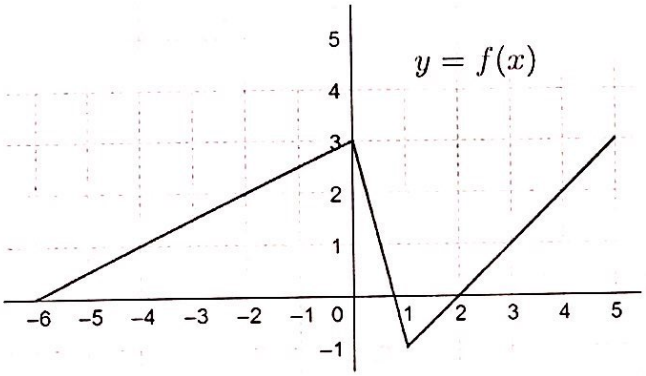
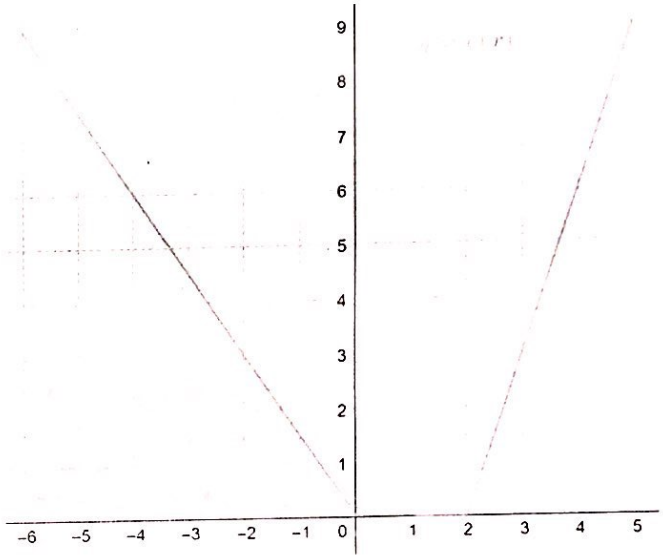
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Math 400 Exam 2 Part 3 40 Total Points

1. The graphs of the function f and g are found below. Let $A(x) = f(x)g(x)$ and $B(x) = \frac{f(x)}{g(x)}$. Use the graphs of f and g to find the following values.

Use exact values. No decimal approximations.





(1.5 pts) $(a.)$ $f'(2)$

$f'(2) = [f(3) - f(1)]/2 \Rightarrow f'(2) = \frac{1+1}{2} = \frac{2}{2} \Rightarrow \boxed{f'(2) = 1}$

(1.5 pts) $(b.)$ $g'(1)$

$g'(1) = (f(2) - f(1))/2 \Rightarrow g'(1) = \frac{0-0}{2} \Rightarrow \boxed{g'(1) = 0}$

(2.5 pts) $(c.)$ $A'(-2)$

$A'(-2) = f(-2)\,g'(-2) + g(-2)\,f'(-2)$

$= 2\left(-\frac{3}{2}\right) + 3\left(\frac{1}{2}\right) = -3 + \frac{3}{2} = -\frac{3}{2} \Rightarrow \boxed{A'(-2) = -\frac{3}{2}}$

(2.5 pts) $(d.)$ $B'(3)$

$B'(3) = \frac{g(3)\,f'(3) - f(3)\,g'(3)}{(g(3))^2}$

$= \frac{3 \times 1 - 1 \times 3}{3^2} = \frac{3-3}{9} = \frac{0}{9} = 0 \Rightarrow \boxed{B'(3) = 0}$

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(8 pts) 2. Find the derivative of the function using the definition of a derivative,

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}.$$

$f(x) = \sqrt{4x+1}$

$$f'(x) = \lim_{h\to 0} \frac{\sqrt{4(x+h)+1} - \sqrt{4x+1}}{h}$$

$$f'(x) = \lim_{h\to 0} \frac{\sqrt{4(x+h)+1} - \sqrt{4x+1}}{h} \left(\frac{\sqrt{4(x+h)+1} + \sqrt{4x+1}}{\sqrt{4(x+h)+1} + \sqrt{4x+1}} \right)$$

$$f'(x) = \lim_{h\to 0} \frac{4(x+h)+1 - (4x+1)}{h\left[\sqrt{4(x+h)+1} + \sqrt{4x+1}\right]}$$

$$f'(x) = \lim_{h\to 0} \frac{4x+4h+1-4x-1}{h\left[\sqrt{4x+4h+1} + \sqrt{4x+1}\right]}$$

$$f'(x) = \lim_{h\to 0} \frac{4h}{h\left[\sqrt{4x+4h+1} + \sqrt{4x+1}\right]}$$

$$f'(x) = \lim_{h\to 0} \frac{4}{\sqrt{4x+4h+1} + \sqrt{4x+1}}$$

$$f'(x) = \frac{4}{\sqrt{4x+1} + \sqrt{4x+1}} \Rightarrow f'(x) = \frac{4}{2\sqrt{4x+1}}$$

$$\boxed{f'(x) = \frac{2}{\sqrt{4x+1}}}$$

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(8 pts) 3. Find the values of x where the tangent line to the graph of f is horizontal. **DO NOT USE LOGARITHMIC DIFFERENTIATION.**

$$h(x) = (2x+3)^7(8x-9)^{12}$$

$$h'(x) = (2x+3)^7 \frac{d}{dx}(8x-9)^{12} + (8x-9)^{12}\frac{d}{dx}(2x+3)^7$$

$$h'(x) = (2x+3)^7\left((12)(8x-9)^{11}(8)\right) + (8x-9)^{12}\left((7)(2x+3)^6(2)\right)$$

$$h'(x) = 96(2x+3)^7(8x-9)^{11} + 14(2x+3)^6(8x-9)^{12}$$

$$h'(x) = 2(2x+3)^6(8x-9)^{11}\left[48(2x+3) + 7(8x-9)\right]$$

f is horizontal which is $h'(x) = 0$

$$2(2x+3)^6(8x-9)^{11}\left[48(2x+3) + 7(8x-9)\right] = 0$$

$(2x+3)^6 = 0$

$2x+3 = 0$

$2x = -3$

$\boxed{x = -\frac{3}{2}}$

$(8x-9)^{11} = 0$

$8x-9 = 0$

$8x = 9$

$\boxed{x = \frac{9}{8}}$

$48(2x+3) + 7(8x-9) = 0$

$96x + 144 + 56x - 63 = 0$

$152x + 81 = 0$

$152x = -81$

$\boxed{x = -\frac{81}{152}}$

The values of x where the tangent line to the graph of f is horizontal are $x = -\frac{3}{2}, \frac{9}{8}, -\frac{81}{152}$.

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(8 pts) 4. Find the derivative of the following function. **DO NOT SIMPLIFY.**

$$y = \log_6 \sqrt[3]{\frac{3x^2+1}{x^4+2}}$$

$$y = \log_6 \left(\frac{3x^2+1}{x^4+2}\right)^{1/3}$$

$$y = \log_6 (3x^2+1)^{1/3} - \log_6 (x^4+2)^{1/3}$$

$$y = \frac{1}{3}\log_6 (3x^2+1) - \frac{1}{3}\log_6 (x^4+2)$$

$$y' = \frac{1}{3}\left(\frac{6x}{(3x^2+1)\ln(6)}\right) - \frac{1}{3}\left(\frac{4x^3}{(x^4+2)\ln(6)}\right)$$

$$\boxed{y' = \frac{2x}{(3x^2+1)\ln(6)} - \frac{4x^3}{3(x^4+2)\ln(6)}}$$

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(8 pts) 5. Find the derivative of the following function. **DO NOT SIMPLIFY.**

$$y = \csc^2[\tan(3^{4x} + \ln x)]$$

$$y = \{\csc[\tan(3^{4x} + \ln x)]\}^2$$

$$y' = 2\csc[\tan(3^{4x} + \ln x)] \cdot \frac{d}{dx}[\csc[\tan(3^{4x} + \ln x)]]$$

$$y' = 2\csc[\tan(3^{4x} + \ln x)] \cdot (-\cot(\tan(3^{4x} + \ln x))\csc(\tan(3^{4x} + \ln x)) \cdot \frac{d}{dx}[\tan[3^{4x} + \ln x]]$$

$$y' = -2\csc^2[\tan(3^{4x} + \ln x)]\cot(\tan(3^{4x} + \ln x))\frac{d}{dx}[\tan(3^{4x} + \ln x)]$$

$$y' = -2\csc^2[\tan(3^{4x} + \ln x)]\cot(\tan(3^{4x} + \ln x)) \cdot \sec^2(3^{4x} + \ln x) \cdot \frac{d}{dx}(3^{4x} + \ln x)$$

$$y' = -2\csc^2[\tan(3^{4x} + \ln x)]\cot(\tan(3^{4x} + \ln x)) \cdot \sec^2(3^{4x} + \ln x) \cdot \left(3^{4x}\ln(3)\cdot 4 + \frac{1}{x}\right)$$

$$y' = -2\csc^2[\tan(3^{4x} + \ln x)]\cot[\tan(3^{4x} + \ln x)] \cdot \sec^2(3^{4x} + \ln x) \cdot \left(\frac{4x\ 3^{4x}\ln(3) + 1}{x}\right)$$

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